

Tokens per Word: our 7 live image prompts

Gemini vs Claude vs GPT-4o · measured 28 September 2026 on the production prompts (main @ c206a39)

Bottom line. Our prompts average **1.44 tokens per word on Gemini** and **1.37 on GPT-4o**, measured with each provider's own tool. The earlier 1.60–1.68 figure was not a measurement: it came from the rough “4 characters = 1 token” shortcut applied to our long words (6.68 characters per word). Claude is estimated: about **1.40** on older models and about **1.81** on the newest ones.

1. Prompt-by-prompt comparison

Prompt	Chars	Words	Google Gemini (exact)		Claude ≤4.6 (estimate)		Claude 4.7+ (estimate)		OpenAI GPT-4o (exact)	
			Tokens	Per word	Tokens	Per word	Tokens	Per word	Tokens	Per word
P1 Clean Studio (₹50 white BG)	6,635	1,027	1,492	1.45	~1,471	1.43	~1,912	1.86	1,450	1.41
P2 Clean E-Commerce	9,746	1,452	2,092	1.44	~2,027	1.40	~2,635	1.81	1,990	1.37
P3 Close-up on Ear	8,898	1,397	1,887	1.35	~1,849	1.32	~2,404	1.72	1,827	1.31
P4 Scale Reference	15,137	2,285	3,314	1.45	~3,186	1.39	~4,142	1.81	3,143	1.38
P5 Professional Studio	12,375	1,753	2,531	1.44	~2,475	1.41	~3,218	1.84	2,405	1.37
P6 Lifestyle	14,025	2,065	3,055	1.48	~2,932	1.42	~3,812	1.85	2,871	1.39
P7 UGC Style	12,216	1,853	2,638	1.42	~2,576	1.39	~3,349	1.81	2,522	1.36
Total / average	79,032	11,832	17,009	1.44	~16,516	1.40	~21,471	1.81	16,208	1.37

How each column was produced. **Gemini:** exact, from Google's free token-counting service (gemini-3.1-flash-image and gemini-3.6-flash gave identical counts). **GPT-4o:** exact, from OpenAI's official tokenizer (o200k_base) run offline. **Claude ≤4.6** (Sonnet 4.6, Haiku 4.5): estimated with OpenAI's previous tokenizer (cl100k_base) as a stand-in, because no Anthropic key is available. **Claude 4.7+:** that estimate plus 30%, per Anthropic's statement that its newer tokenizer produces “approximately 30% more tokens”. **Words:** anything separated by spaces, so bullet dashes count as words. P1 is the ₹50 single shot; P2–P7 are the six E-Com Pack 1 shots.

2. Why we are above the usual 1.30 rule of thumb

- The 1.30 rule describes everyday English, where most words are short and common, like “the” or “photo”. The AI stores those as a single token each.
- Our prompts read like a technical rulebook: long specialist words, CAPITAL headings and hundreds of bullet points. Those get split into more pieces, which puts us at about 1.4 tokens per word. That is only moderately above the rule.
- The 1.60–1.68 figure was a by-product of the shortcut: 6.68 characters per word ÷ 4 = 1.67. In reality Gemini fits about 4.65 of our characters into each token, so it uses fewer tokens than the shortcut predicted.
- Claude's newest models split text more finely and will land near 1.8 tokens per word on the same prompts.

3. What formatting and jewellery terms add

Feature	What happens	Extra tokens (all 7 prompts)	Share
Bullet points and numbering (742 lines)	Each “- ” or “1.” marker, and each line break, costs a token that is not a real word	+656 markers, +331 line breaks	≈ 6%
CAPITALS (867 words)	Capitals are rarer, so they are split: “ABSOLUTE” = 3 tokens vs “absolute” = 1; “MUST” = 2	+425	2.6%
Hyphenated jewellery terms (201)	The hyphen breaks the word: “micro-pave” = micro -p ave (3); “lever-backs” = 3; “non-negotiable” = 4	+114	0.7%
Specialist / British spellings	“jewellery” = 3 tokens, “filigree” = 3, “earring” = 2	spread throughout	—
Hex codes (#FFFFFF, 11×)	“#FFFFFF” is only 2 tokens, but “RGB(255, 255, 255)” is 10 because every comma and number is its own piece	+5	≈ 0%
Other punctuation	Colons, brackets and dashes are usually 1 token each	+50	0.3%

Measured exactly on GPT-4o by removing each feature and recounting. Overall, about **24%** of our tokens are pure punctuation, spaces or numbers rather than words.

4. What it means for cost

All of this formatting together adds roughly 1,500 tokens across the 7 prompts, about **₹0.07 in total** on Gemini (input text at \$0.50 per million tokens, ₹95.93 per US\$). The whole prompt text for a ₹50 single shot costs about ₹0.10; the finished image is about 97% of the AI bill. Trimming prompt formatting to save money is not worthwhile, and the bullets and capitals may help the AI follow the rules. Real cost control comes from limiting repeat generations and fallback use, not prompt length.

Sources: Gemini count_tokens API; OpenAI tiktoken (o200k_base, cl100k_base); Anthropic pricing documentation on the Claude 4.7+ tokenizer; Gemini API pricing page (ai.google.dev/gemini-api/docs/pricing). Exact Claude counts can be obtained free via Anthropic's token-counting endpoint once an API key is added.